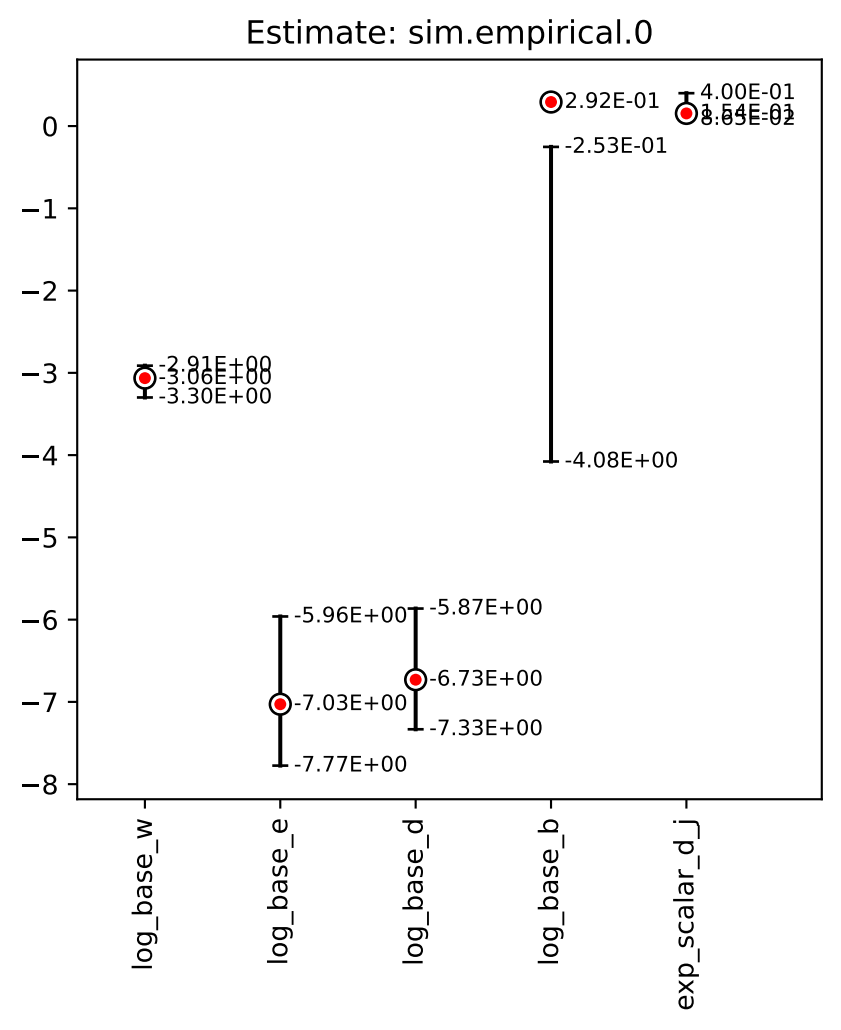
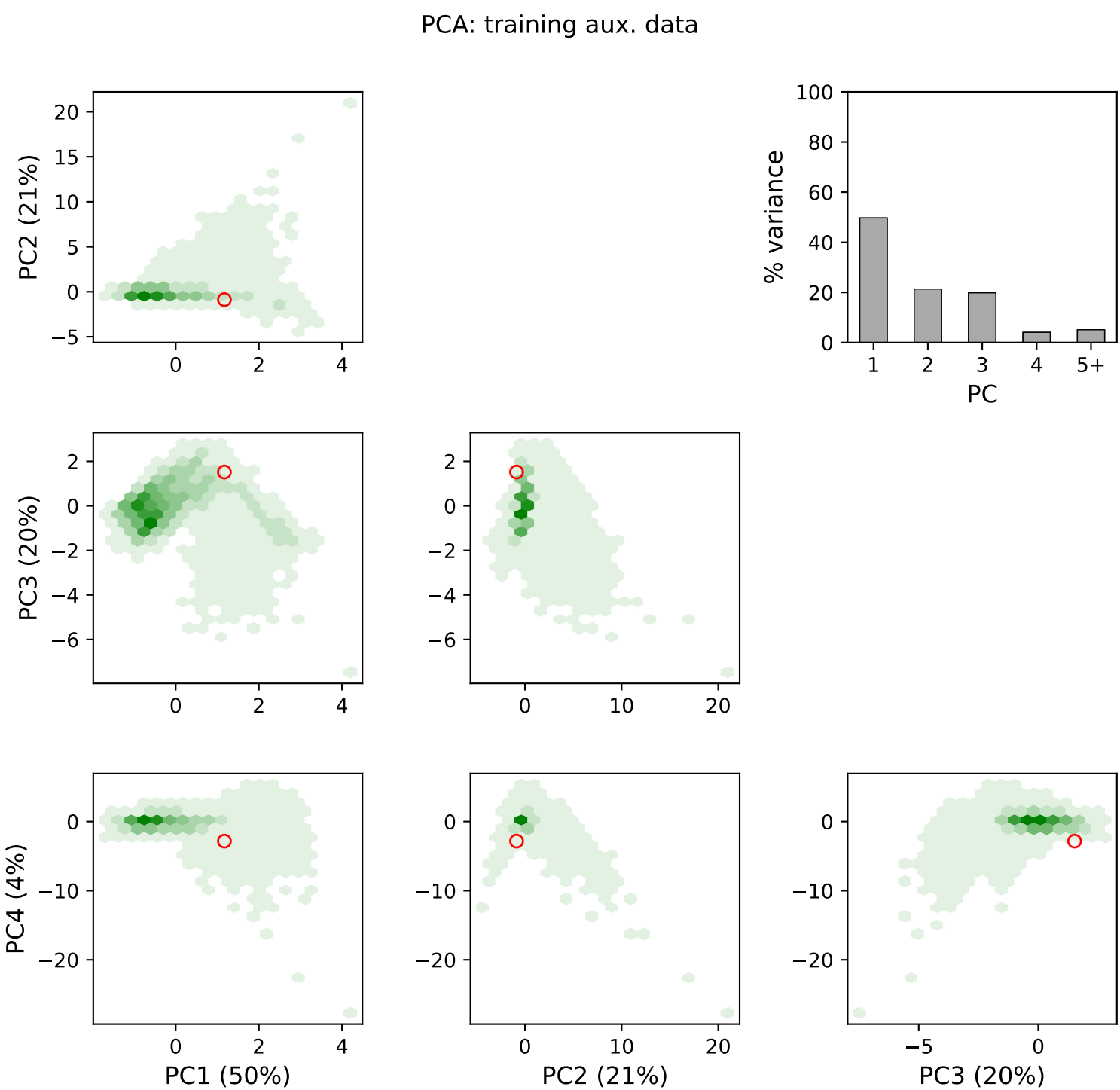
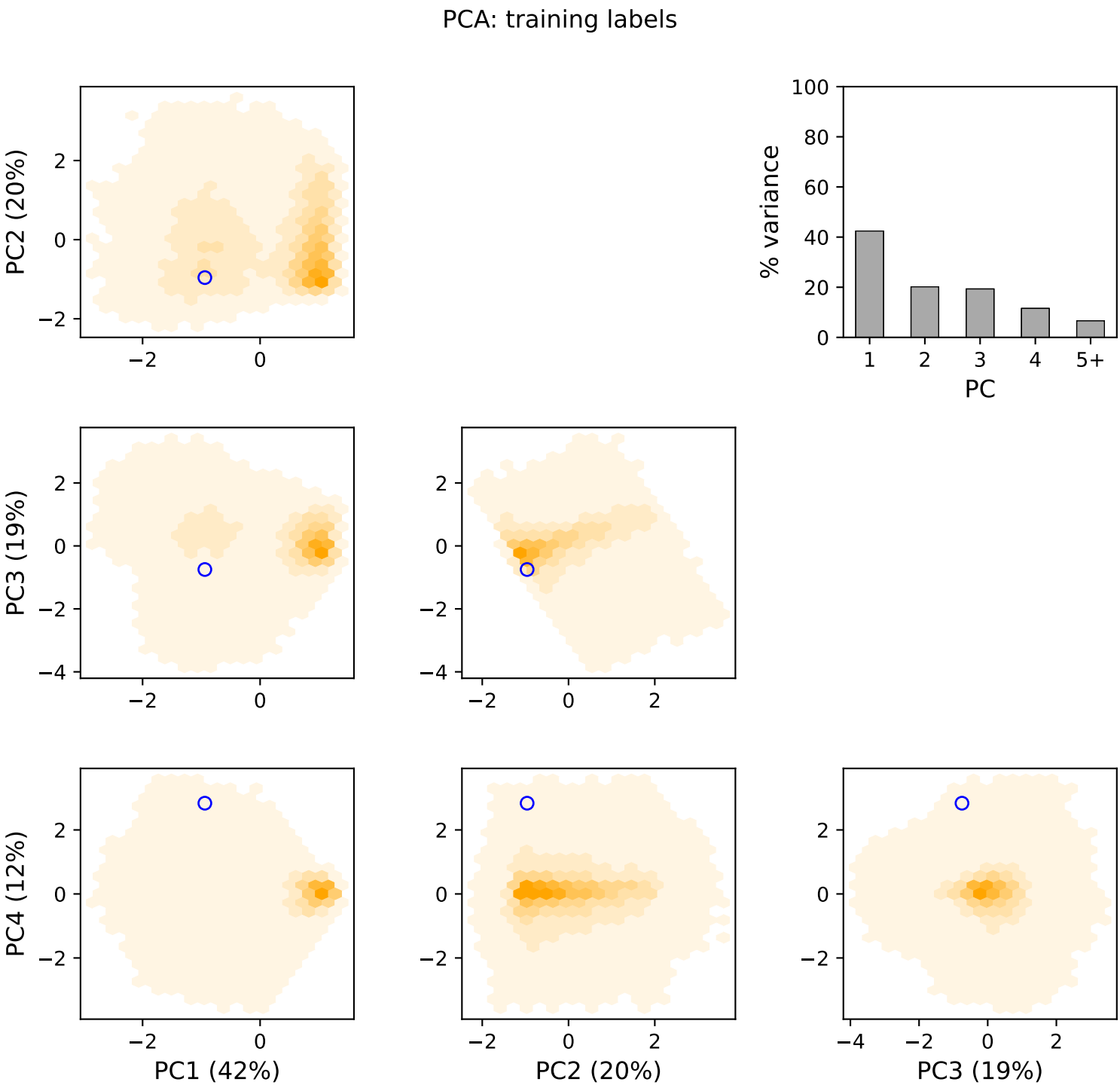
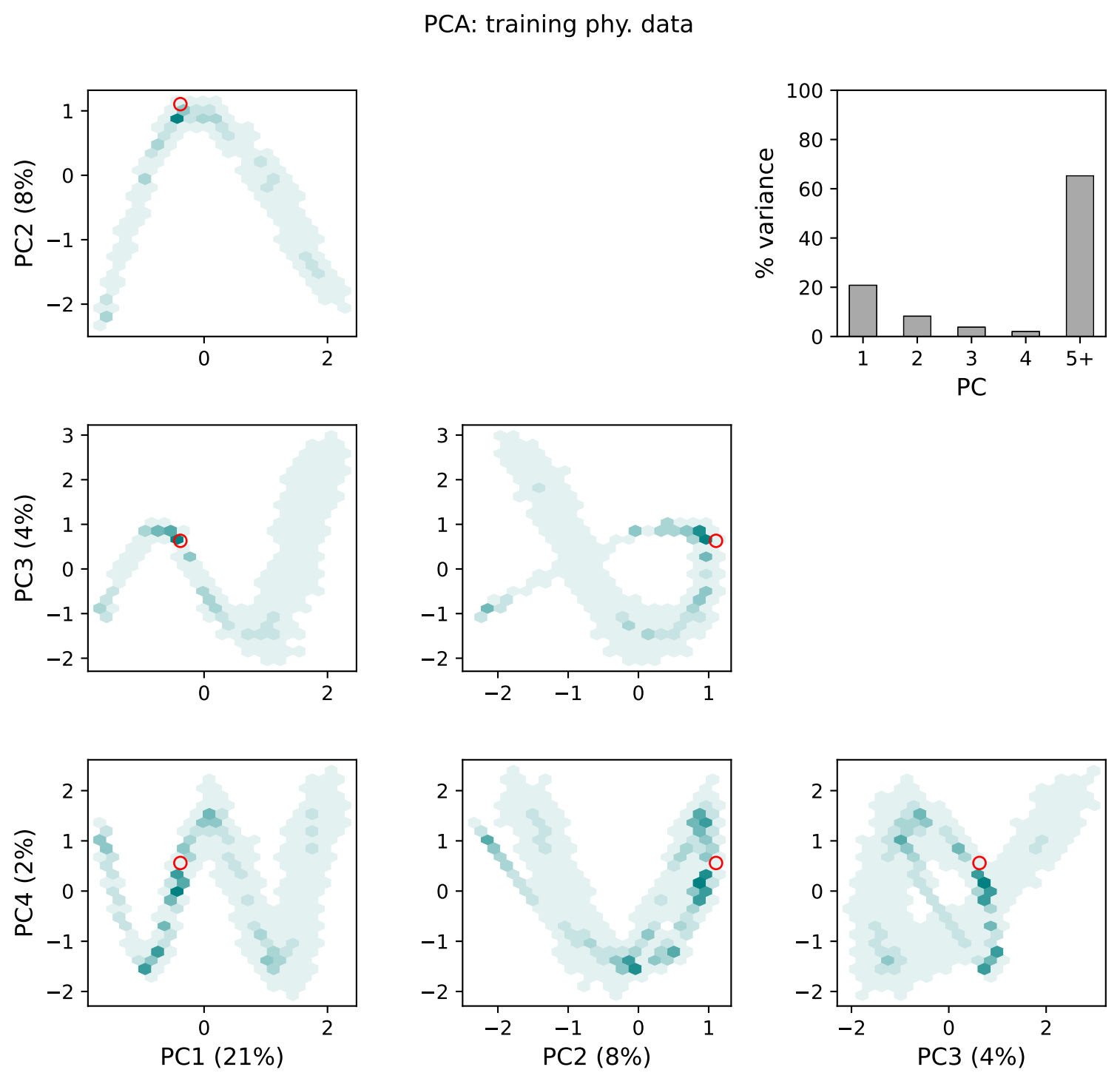
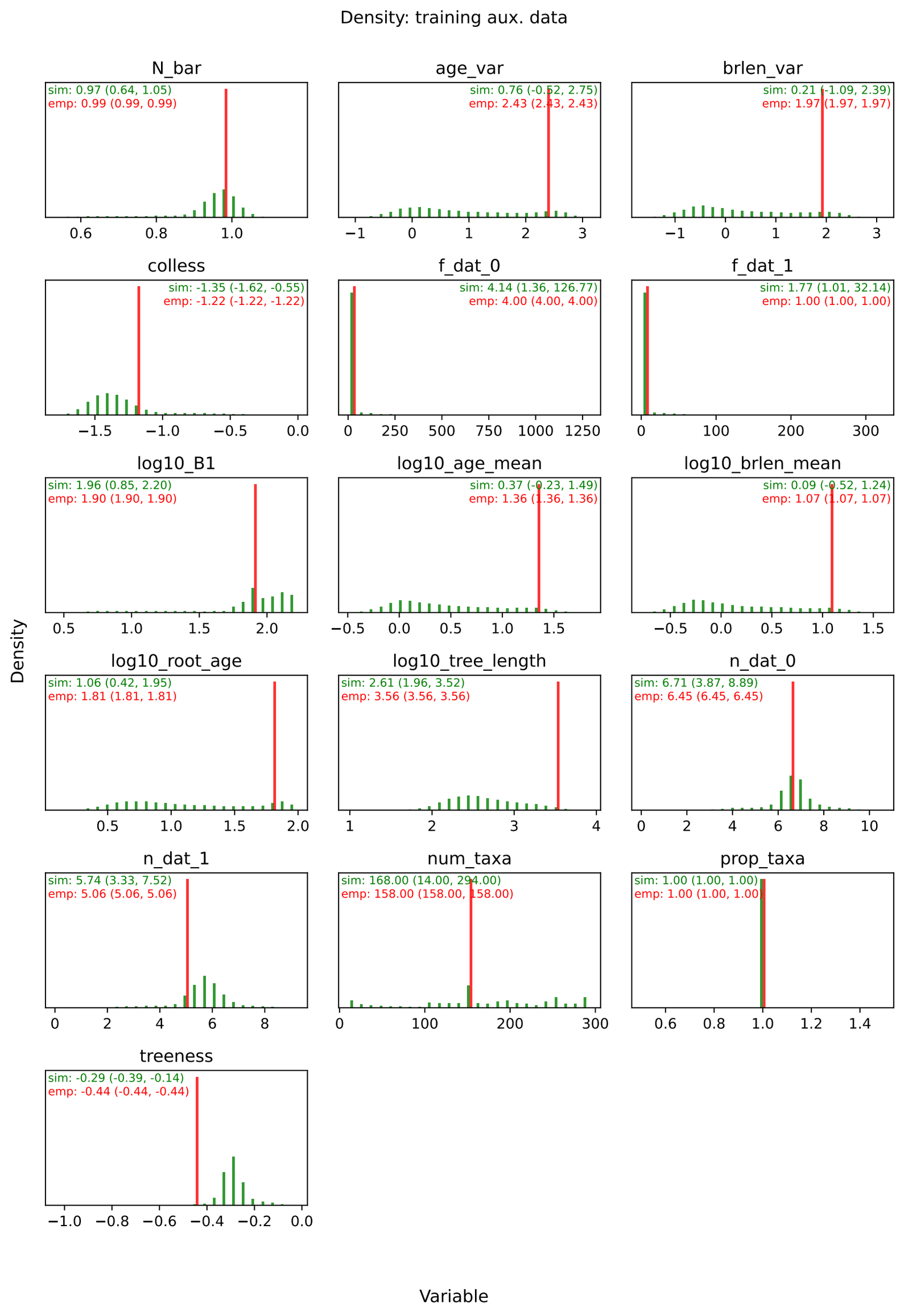
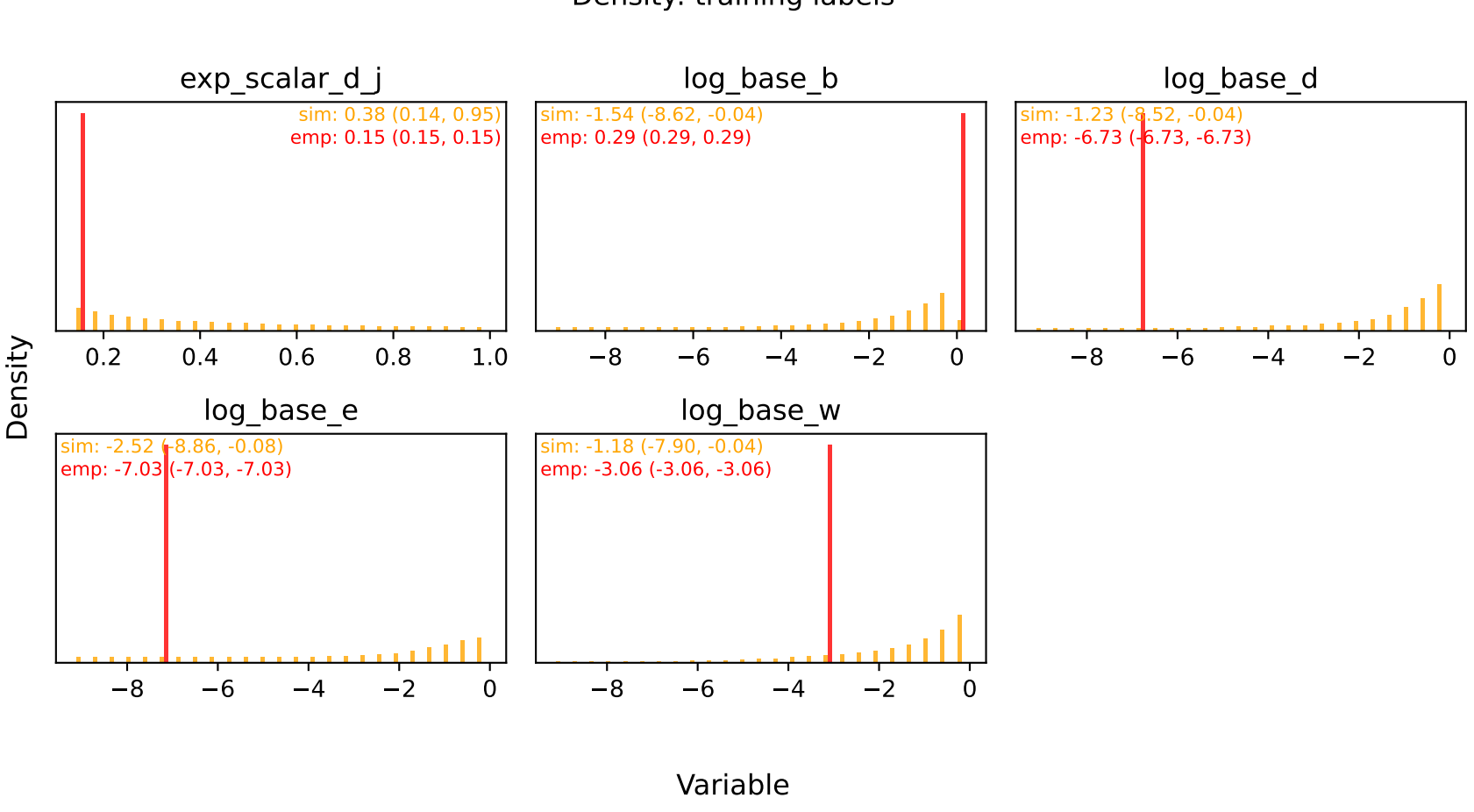

Estimate: sim.empirical.0
0
−1
−2
−3
−4
−5
−6
−7
−8
-2.91E+00
-3.06E+00
-3.30E+00
-5.96E+00
-7.03E+00
-7.77E+00
-5.87E+00
-6.73E+00
-7.33E+00
2.92E-01
-2.53E-01
-4.08E+00
4.00E-01
log_base_w
log_base_e
log_base_d
log_base_b
exp_scalar_d_j

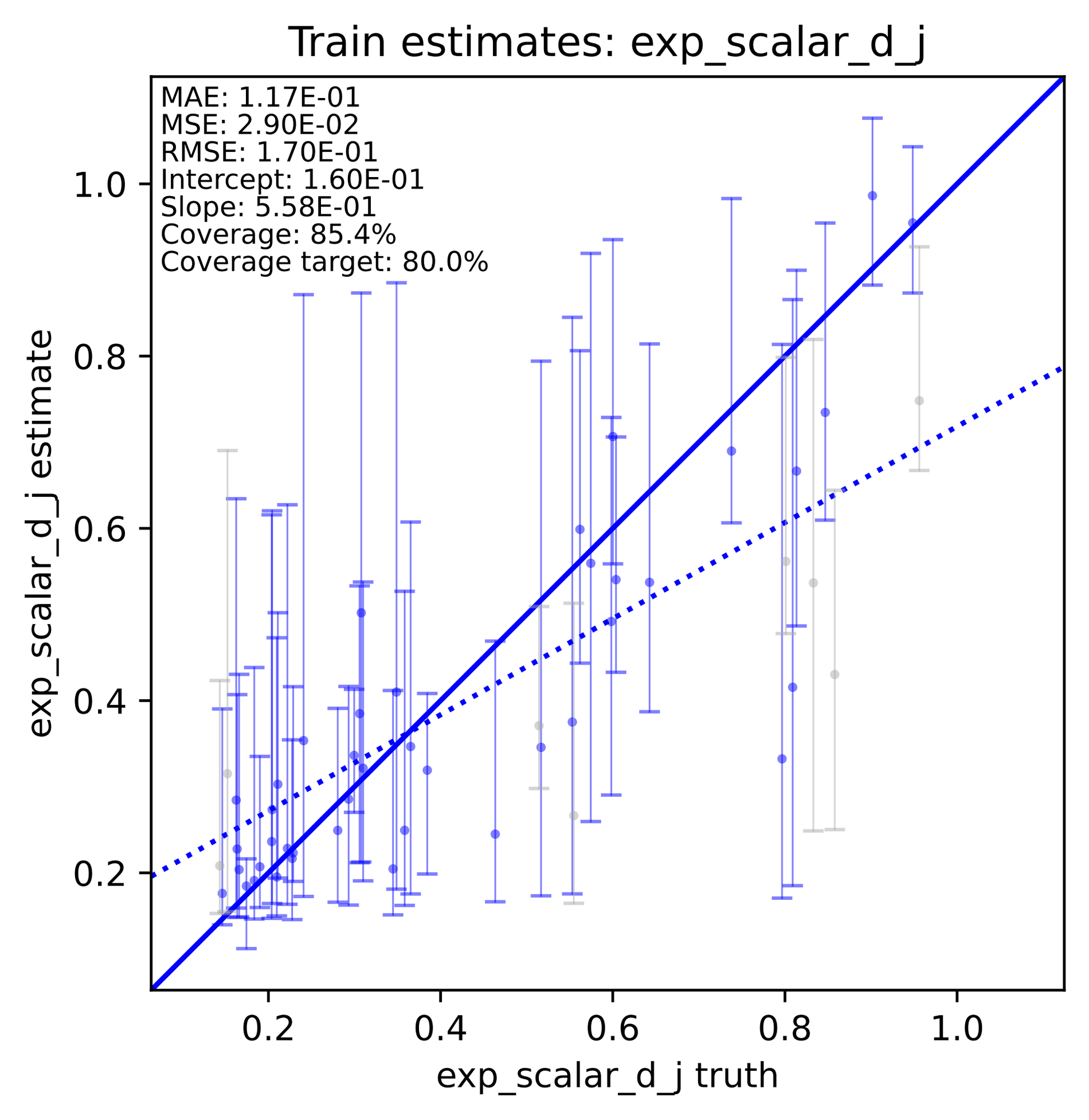

PCA: training aux. data
PC2 (21%)
% variance
PC
PC3 (20%)
PC4 (4%)
PC1 (50%)
PC2 (21%)
PC3 (20%)

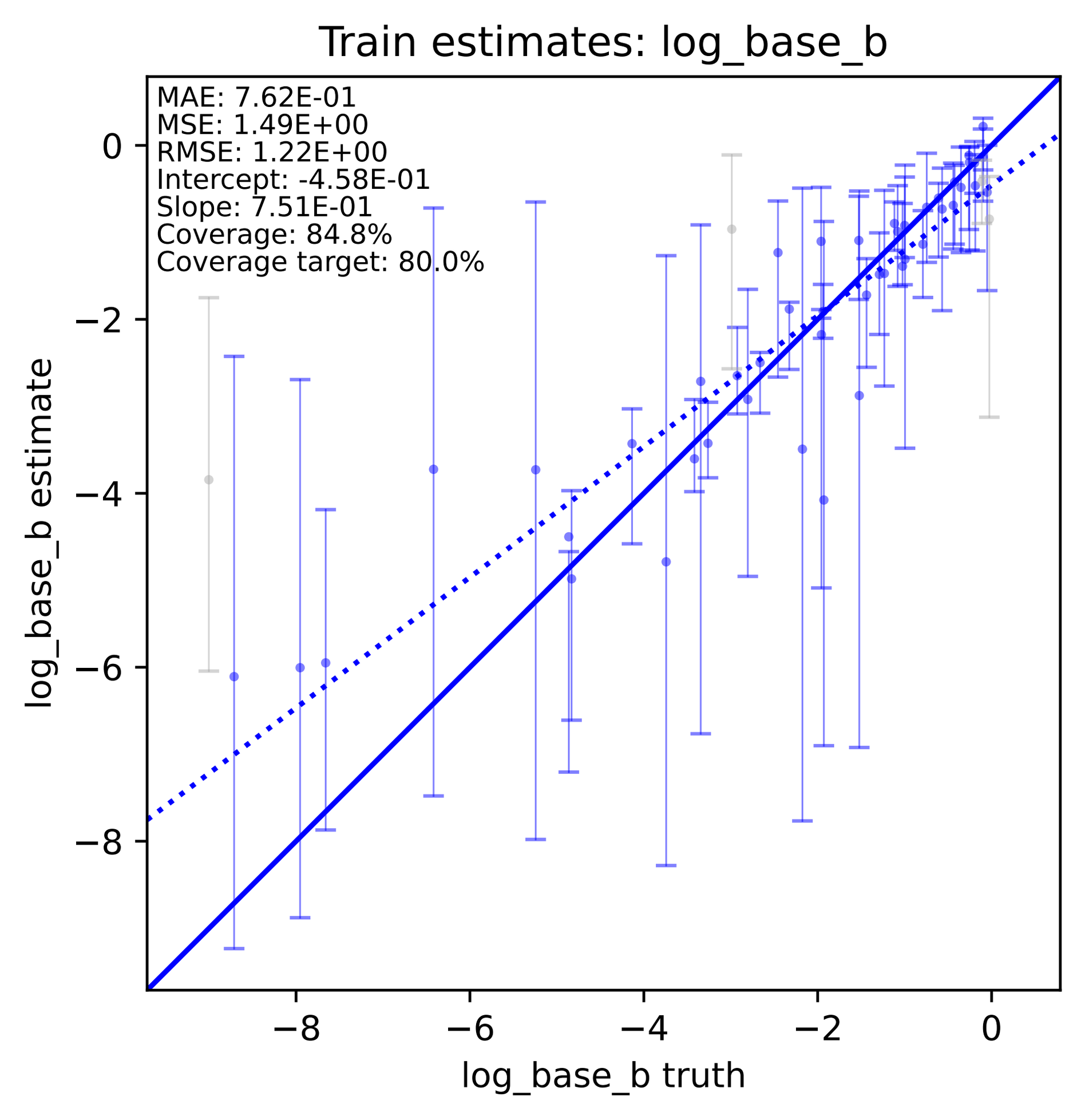

PCA: training labels
PC2 (20%)
% variance
100
80
60
40
20
0
1
2
3
4
5+
PC
PC3 (19%)
PC4 (12%)
PC1 (42%)
PC2 (20%)
PC3 (19%)

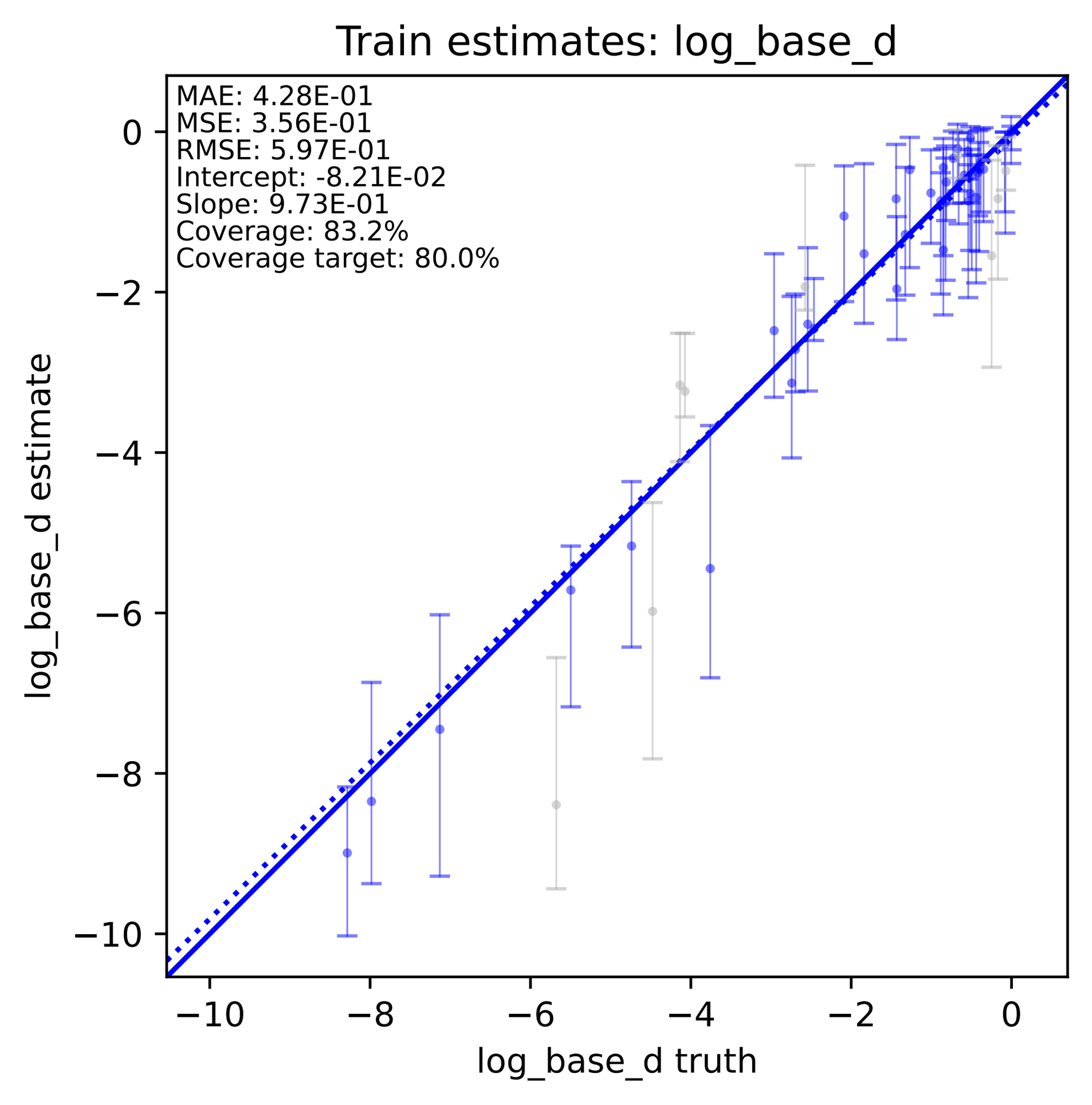

PCA: training phy. data
PC2 (8%)
% variance
PC
PC3 (4%)
PC4 (2%)
PC1 (21%)
PC2 (8%)
PC3 (4%)

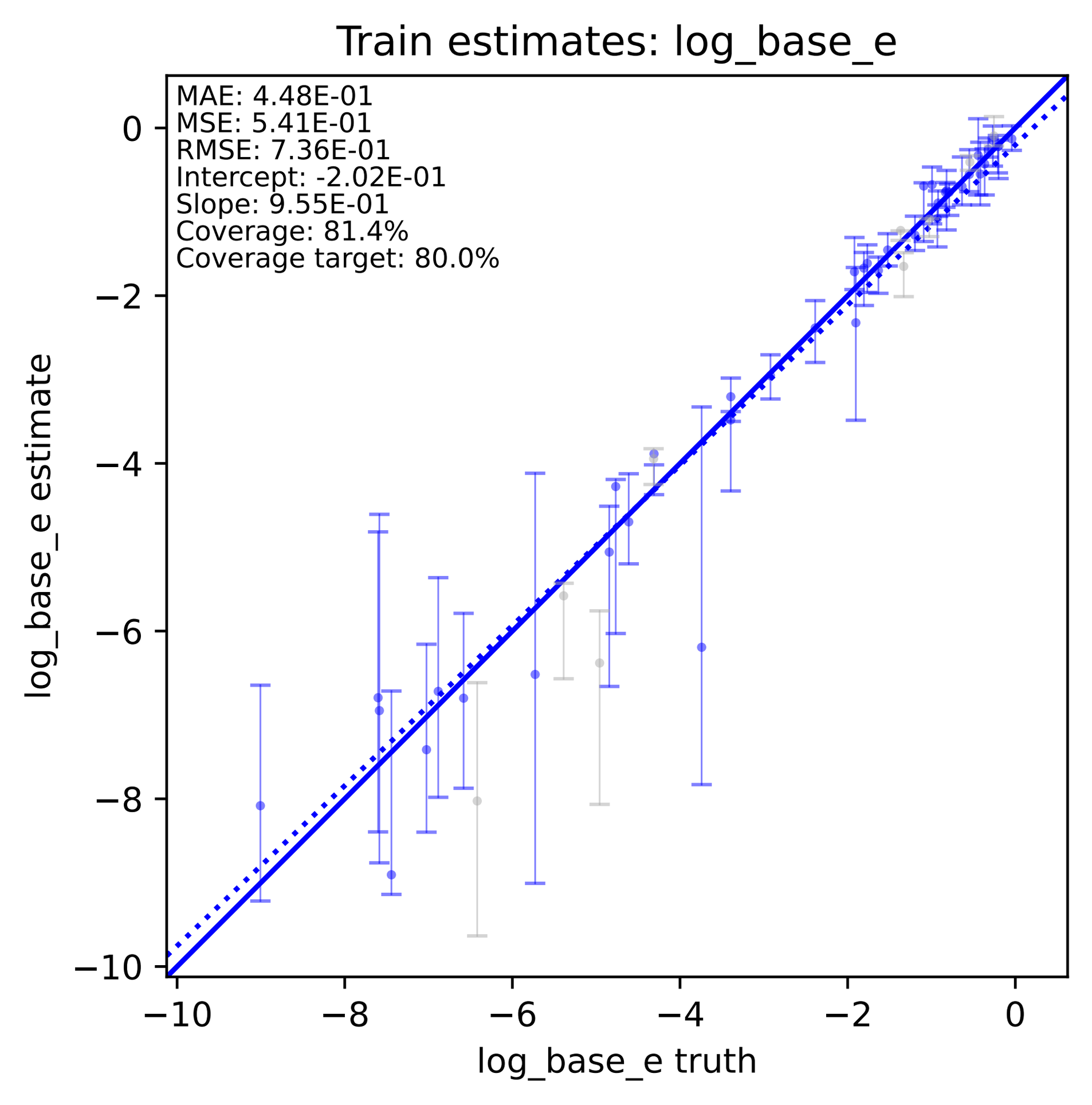

Density: training aux. data
N_bar
sim: 0.97 (0.64, 1.05)
emp: 0.99 (0.99, 0.99)
age_var
sim: 0.76 (-0.52, 2.75)
emp: 2.43 (2.43, 2.43)
brlen_var
sim: 0.21 (-1.09, 2.39)
emp: 1.97 (1.97, 1.97)
colless
sim: -1.35 (-1.62, -0.55)
emp: -1.22 (-1.22, -1.22)
f_dat_0
sim: 4.14 (1.36, 126.77)
emp: 4.00 (4.00, 4.00)
f_dat_1
sim: 1.77 (1.01, 32.14)
emp: 1.00 (1.00, 1.00)
log10_B1
sim: 1.96 (0.85, 2.20)
emp: 1.90 (1.90, 1.90)
log10_age_mean
sim: 0.37 (-0.23, 1.49)
emp: 1.36 (1.36, 1.36)
log10_brlen_mean
sim: 0.09 (-0.52, 1.24)
emp: 1.07 (1.07, 1.07)
log10_root_age
sim: 1.06 (0.42, 1.95)
emp: 1.81 (1.81, 1.81)
log10_tree_length
sim: 2.61 (1.96, 3.52)
emp: 3.56 (3.56, 3.56)
n_dat_0
sim: 6.71 (3.87, 8.89)
emp: 6.45 (6.45, 6.45)
n_dat_1
sim: 5.74 (3.33, 7.52)
emp: 5.06 (5.06, 5.06)
num_taxa
sim: 168.00 (14.00, 294.00)
emp: 158.00 (158.00, 158.00)
prop_taxa
sim: 1.00 (1.00, 1.00)
emp: 1.00 (1.00, 1.00)
treeness
sim: -0.29 (-0.39, -0.14)
emp: -0.44 (-0.44, -0.44)
Density
Variable

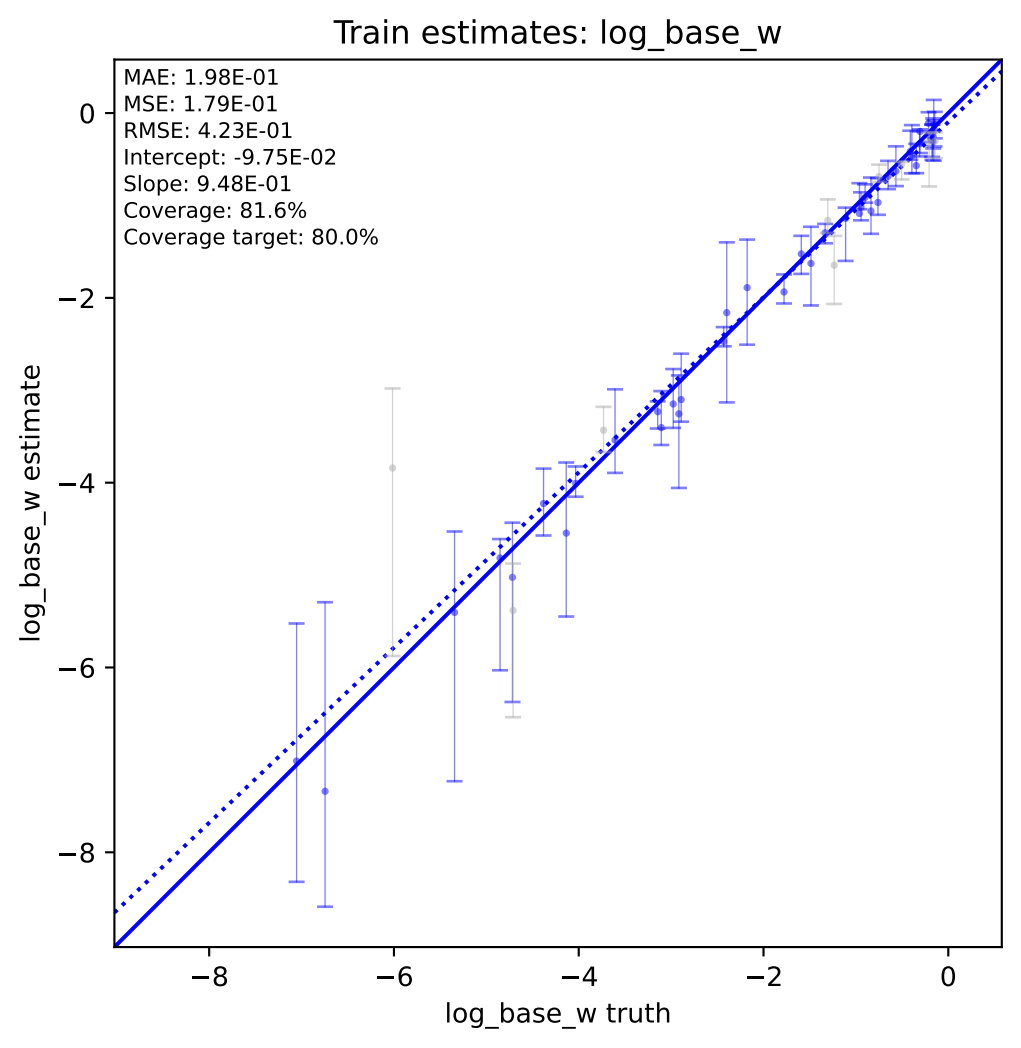

Density: training labels
exp_scalar_d_j
sim: 0.38 (0.14, 0.95)
emp: 0.15 (0.15, 0.15)
log_base_b
sim: -1.54 (-8.62, -0.04)
emp: 0.29 (0.29, 0.29)
log_base_d
sim: -1.23 (-8.52, -0.04)
emp: -6.73 (-6.73, -6.73)
log_base_e
sim: -2.52 (-8.86, -0.08)
emp: -7.03 (-7.03, -7.03)
log_base_w
sim: -1.18 (-7.90, -0.04)
emp: -3.06 (-3.06, -3.06)
Density
Variable

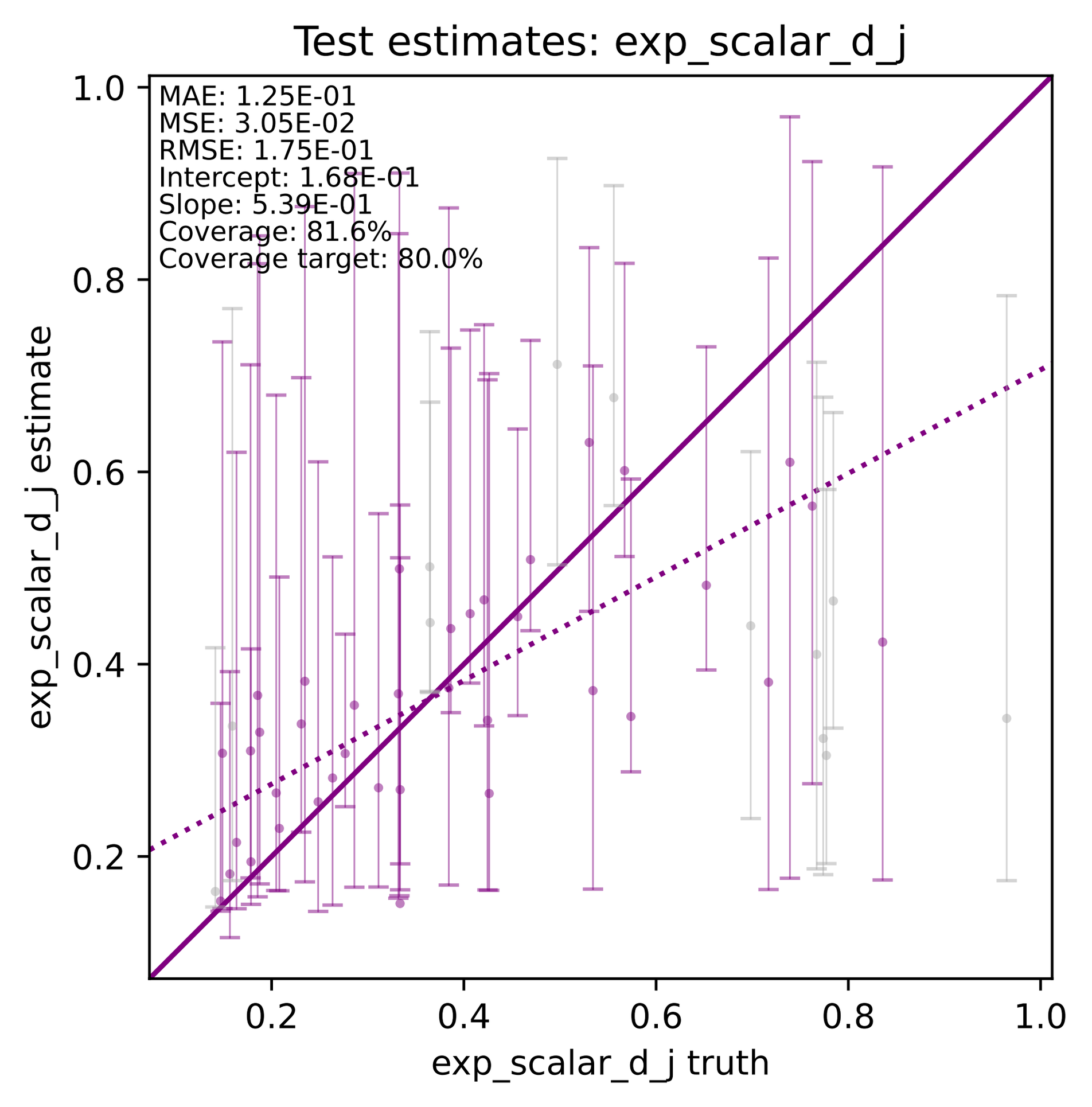

Train estimates: exp_scalar_d_j
MAE: 1.17E-01
MSE: 2.90E-02
RMSE: 1.70E-01
Intercept: 1.60E-01
Slope: 5.58E-01
Coverage: 85.4%
Coverage target: 80.0%
exp_scalar_d_j estimate
exp_scalar_d_j truth
0.2
0.4
0.6
0.8
1.0

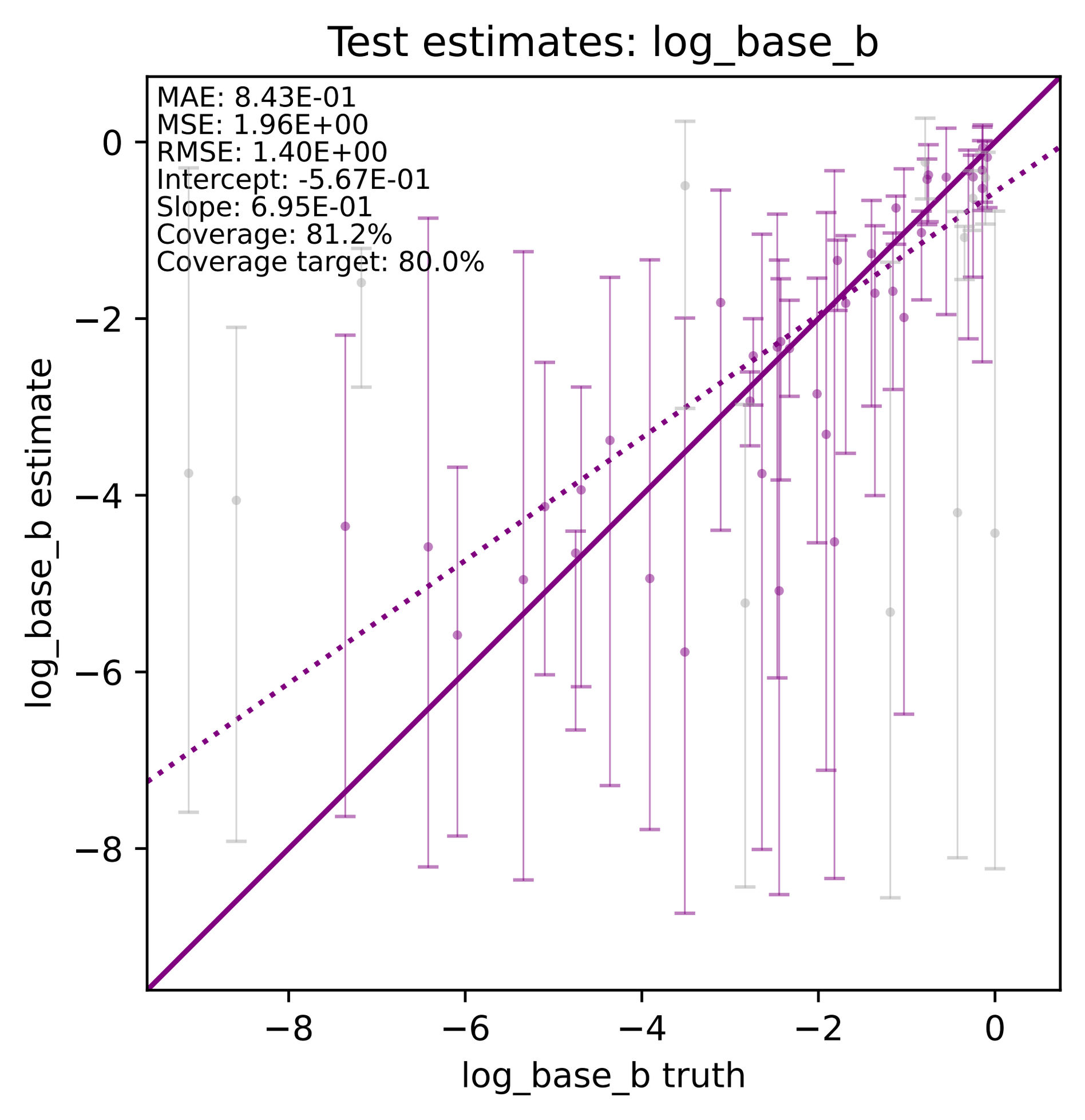

Train estimates: log_base_b
MAE: 7.62E-01
MSE: 1.49E+00
RMSE: 1.22E+00
Intercept: -4.58E-01
Slope: 7.51E-01
Coverage: 84.8%
Coverage target: 80.0%
log_base_b estimate
log_base_b truth
0
−2
−4
−6
−8
−8
−6
−4
−2
0

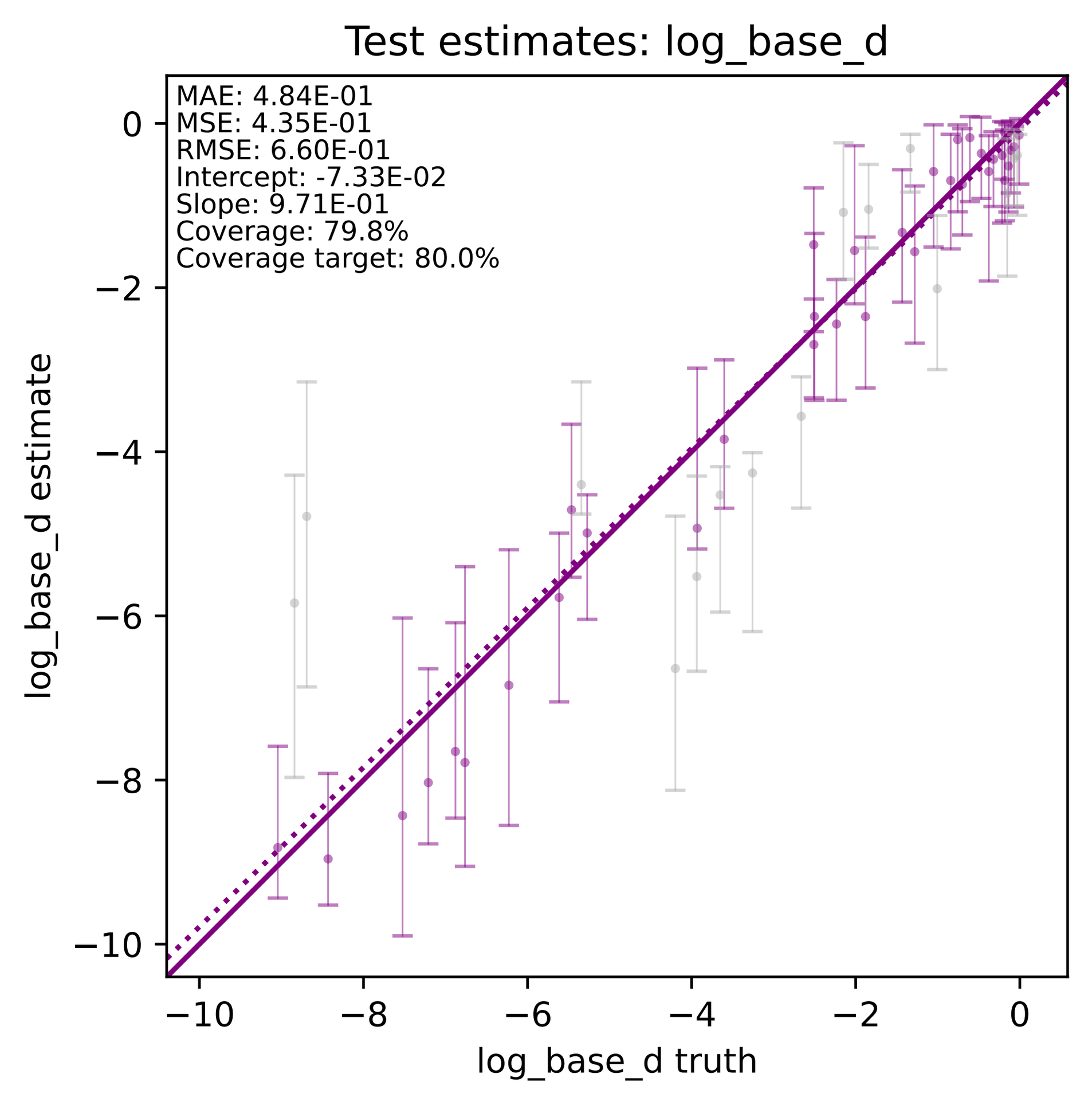

Train estimates: log_base_d
MAE: 4.28E-01
MSE: 3.56E-01
RMSE: 5.97E-01
Intercept: -8.21E-02
Slope: 9.73E-01
Coverage: 83.2%
Coverage target: 80.0%
log_base_d estimate
log_base_d truth
0
−2
−4
−6
−8
−10
−10
−8
−6
−4
−2
0

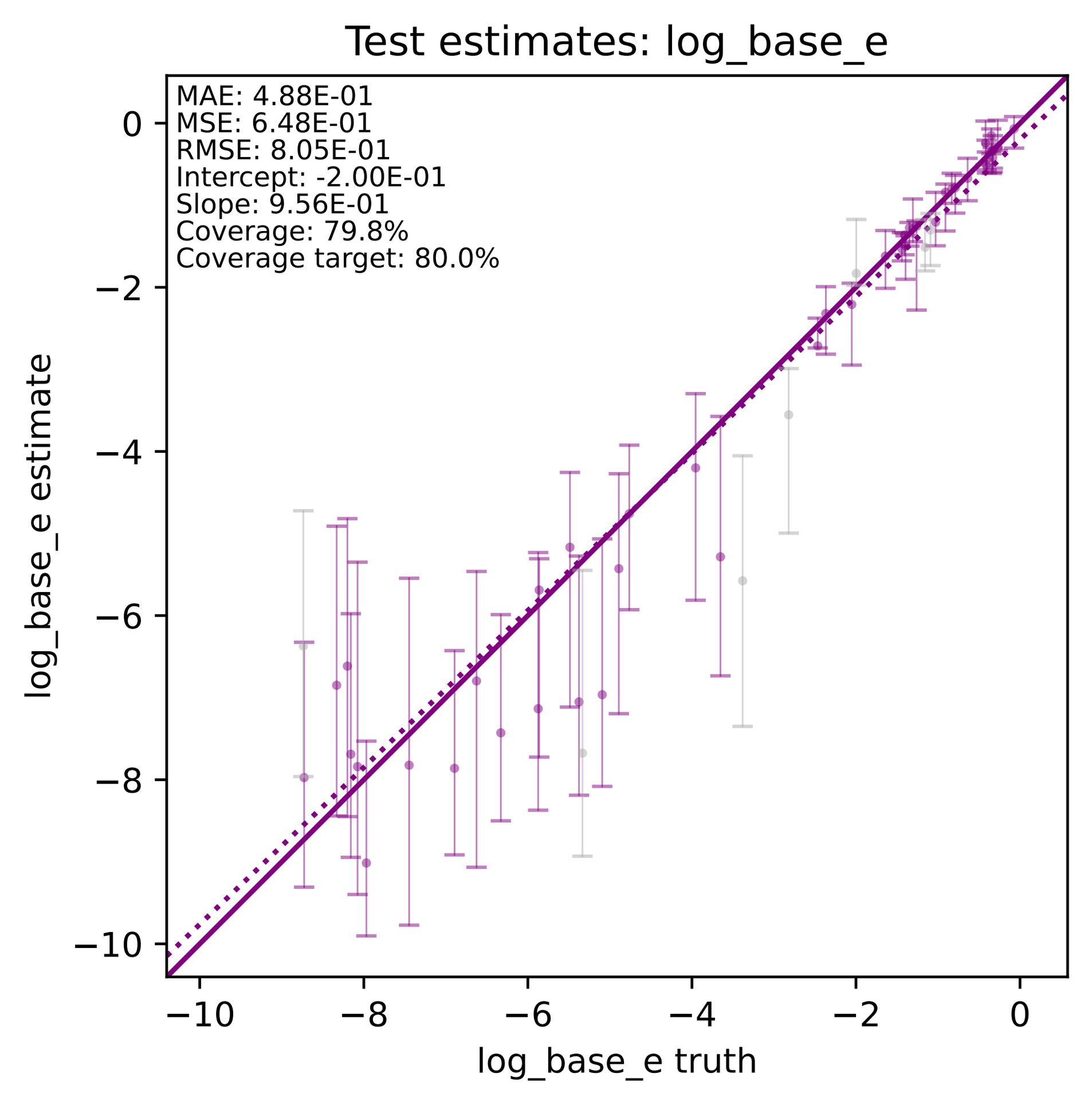

Train estimates: log_base_e
MAE: 4.48E-01
MSE: 5.41E-01
RMSE: 7.36E-01
Intercept: -2.02E-01
Slope: 9.55E-01
Coverage: 81.4%
Coverage target: 80.0%
log_base_e estimate
log_base_e truth
0
−2
−4
−6
−8
−10
−10
−8
−6
−4
−2
0

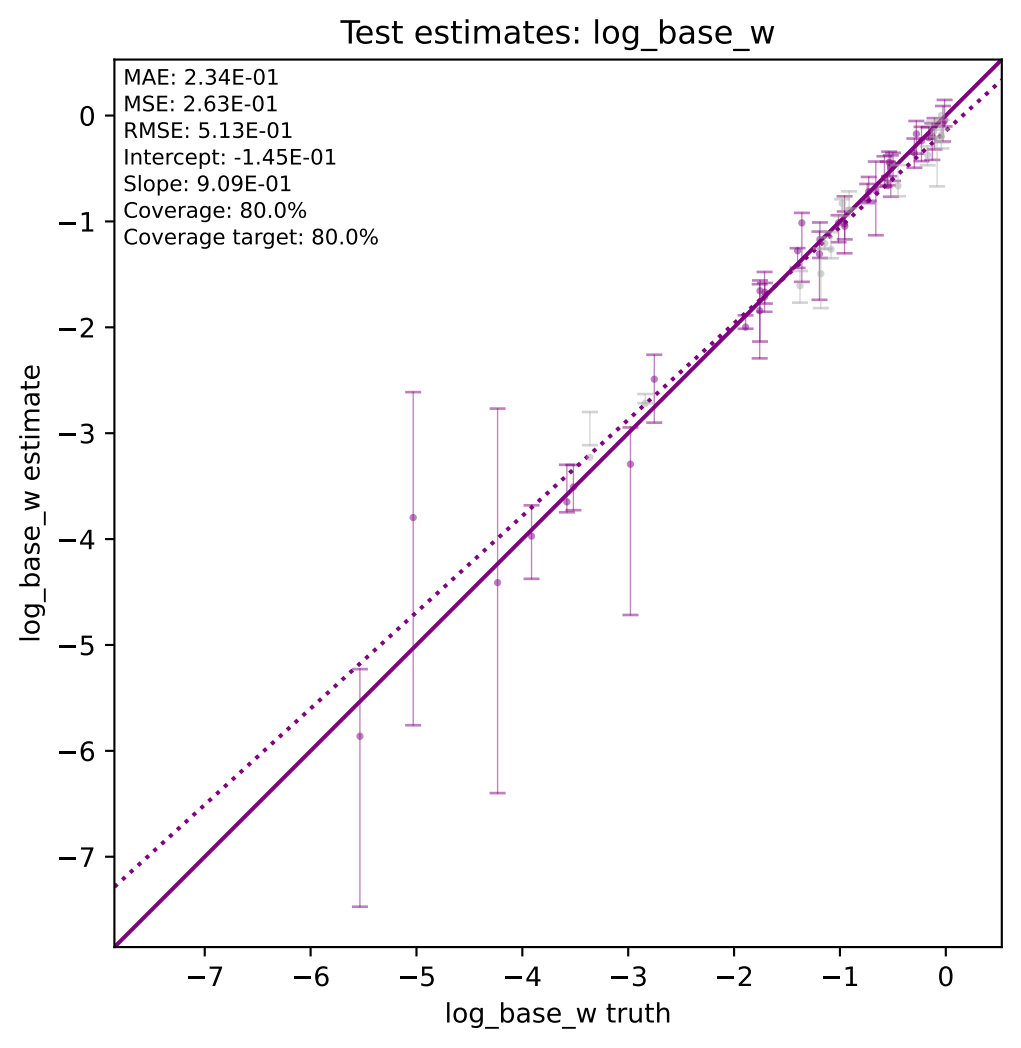

Train estimates: log_base_w
MAE: 1.98E-01
MSE: 1.79E-01
RMSE: 4.23E-01
Intercept: -9.75E-02
Slope: 9.48E-01
Coverage: 81.6%
Coverage target: 80.0%
log_base_w estimate
log_base_w truth
0
−2
−4
−6
−8
−8
−6
−4
−2
0

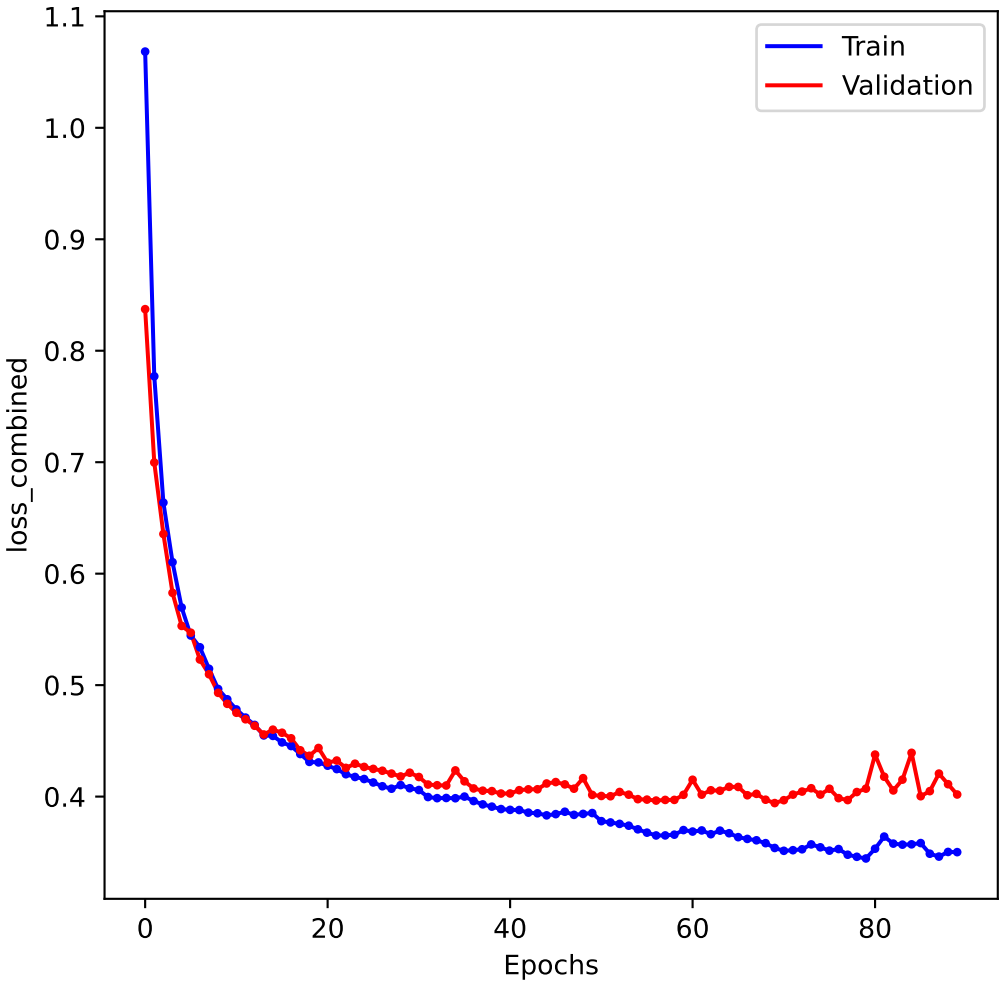

Test estimates: exp_scalar_d_j
MAE: 1.25E-01
MSE: 3.05E-02
RMSE: 1.75E-01
Intercept: 1.68E-01
Slope: 5.39E-01
Coverage: 81.6%
Coverage target: 80.0%
exp_scalar_d_j estimate
exp_scalar_d_j truth
1.0
0.8
0.6
0.4
0.2
0.2
0.4
0.6
0.8
1.0

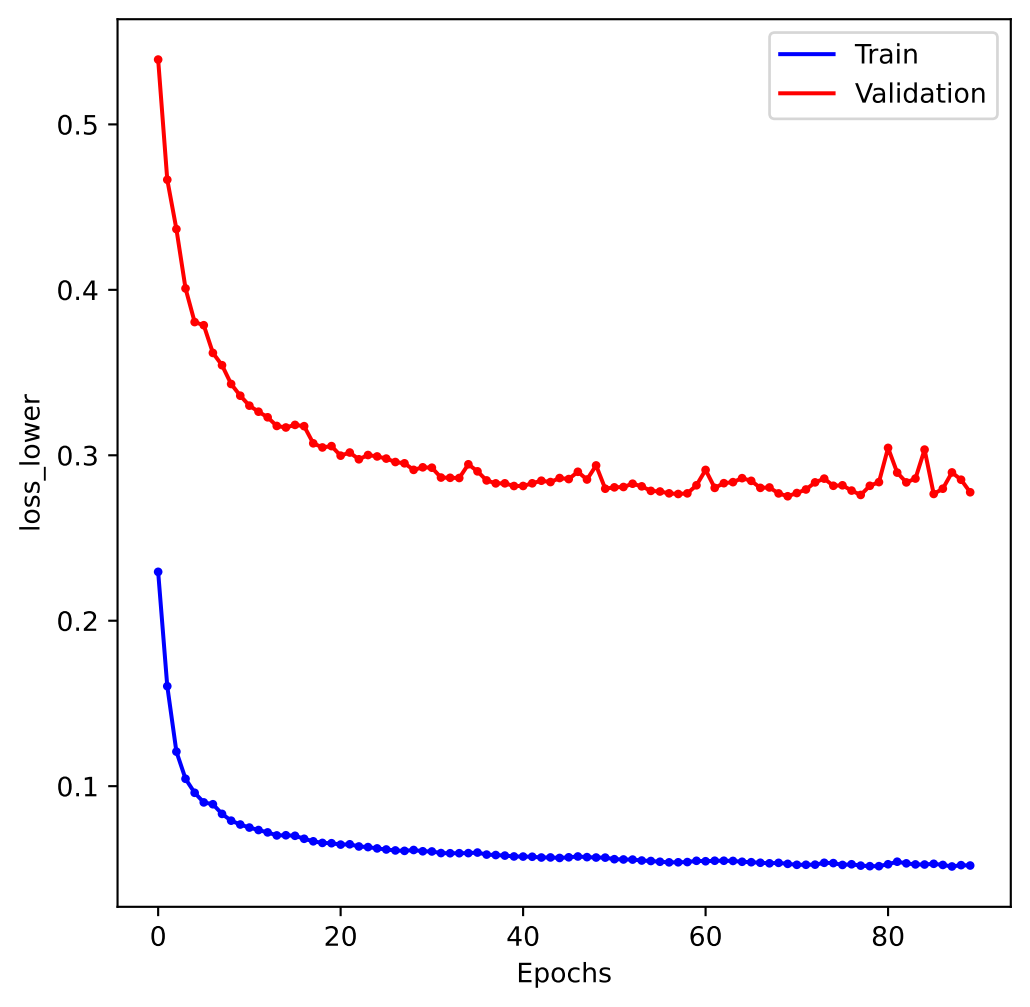

Test estimates: log_base_b
MAE: 8.43E-01
MSE: 1.96E+00
RMSE: 1.40E+00
Intercept: -5.67E-01
Slope: 6.95E-01
Coverage: 81.2%
Coverage target: 80.0%
log_base_b estimate
log_base_b truth
0
−2
−4
−6
−8
−8
−6
−4
−2
0

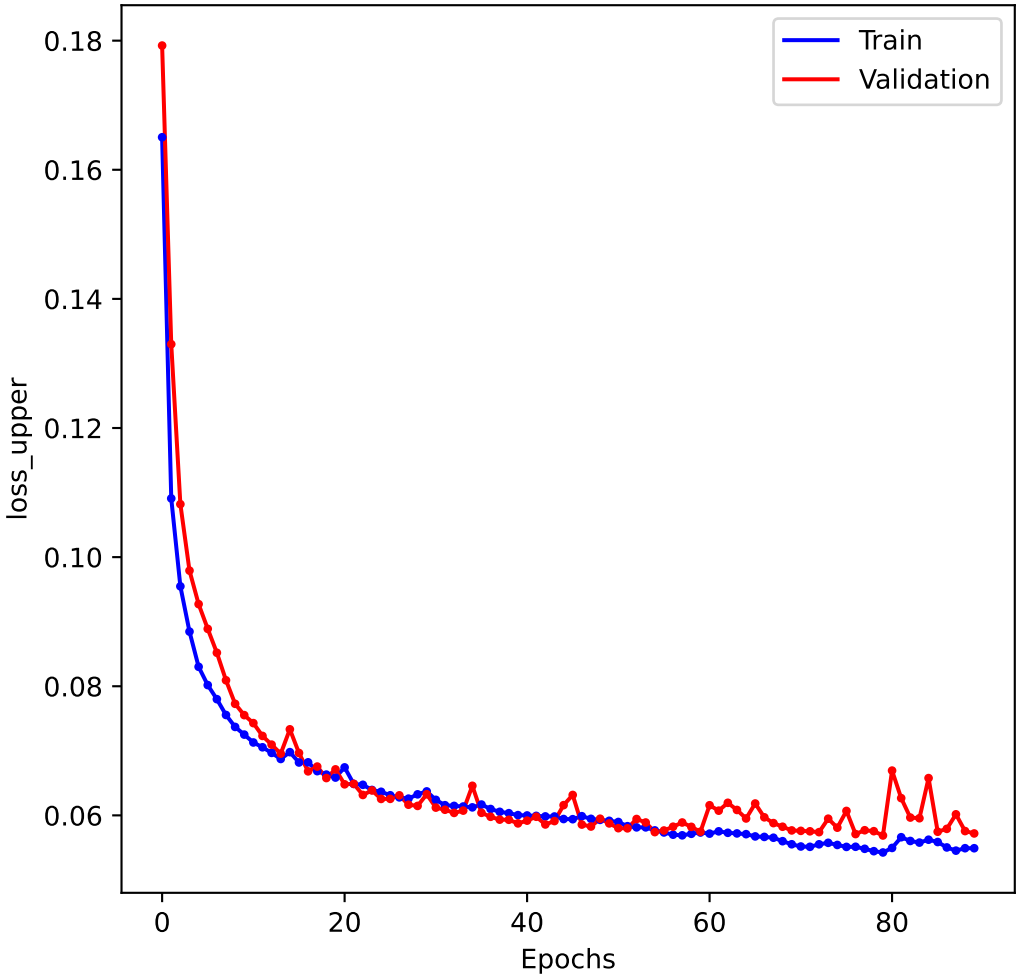

Test estimates: log_base_d
MAE: 4.84E-01
MSE: 4.35E-01
RMSE: 6.60E-01
Intercept: -7.33E-02
Slope: 9.71E-01
Coverage: 79.8%
Coverage target: 80.0%
log_base_d estimate
log_base_d truth
0
−2
−4
−6
−8
−10
−10
−8
−6
−4
−2
0

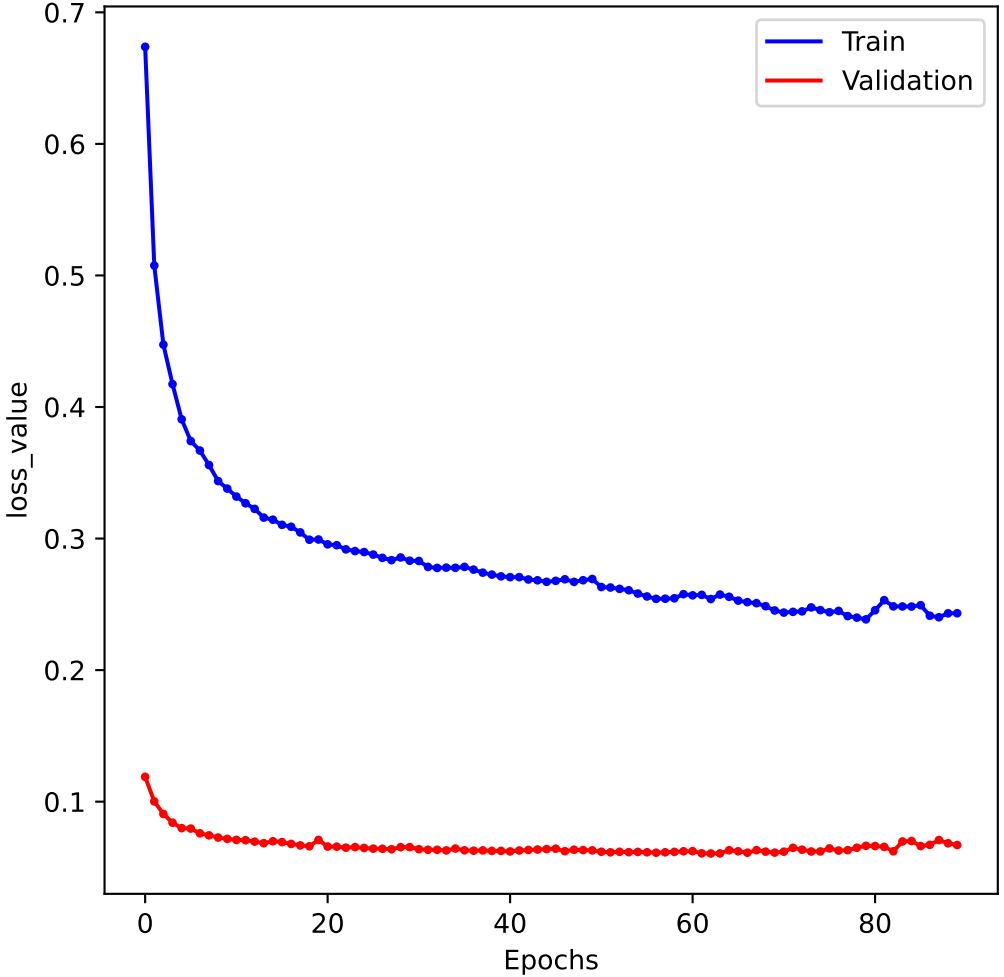

Test estimates: log_base_e
MAE: 4.88E-01
MSE: 6.48E-01
RMSE: 8.05E-01
Intercept: -2.00E-01
Slope: 9.56E-01
Coverage: 79.8%
Coverage target: 80.0%
log_base_e estimate
log_base_e truth
0
−2
−4
−6
−8
−10
−10
−8
−6
−4
−2
0

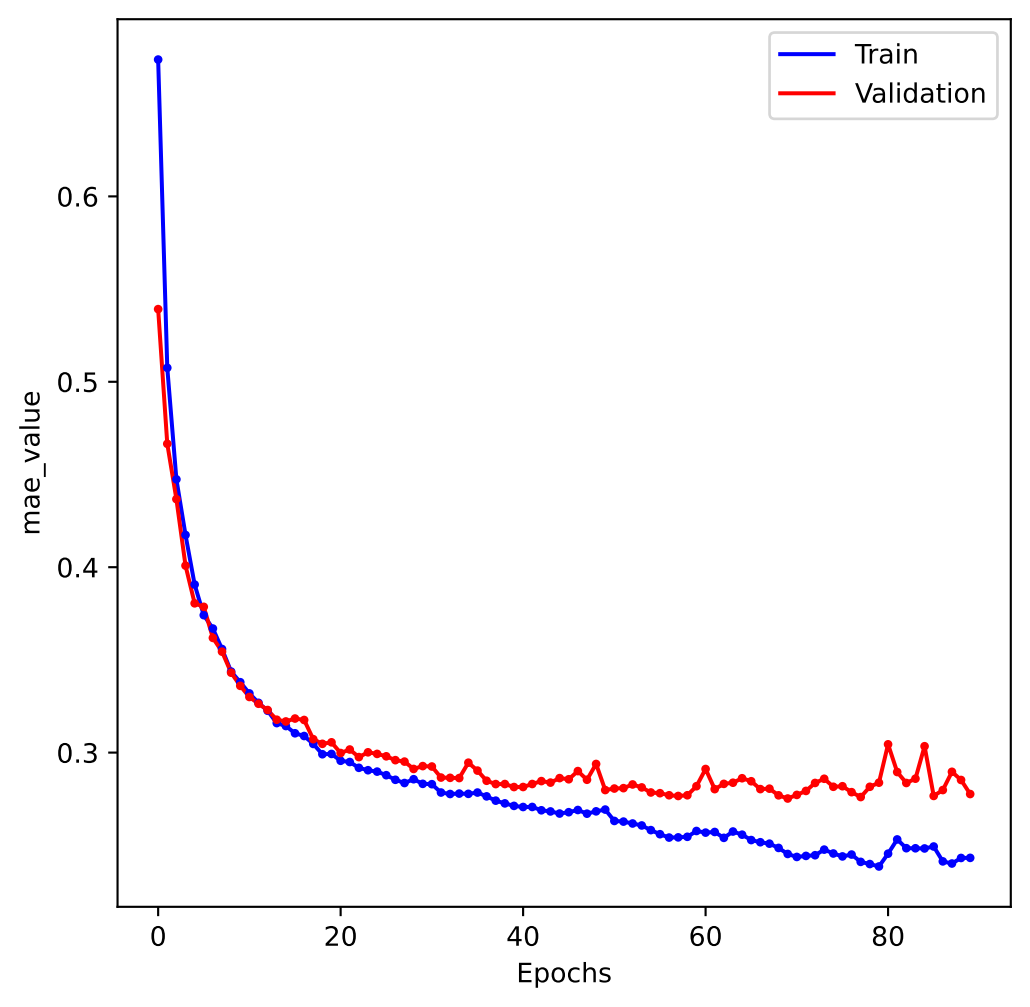

Test estimates: log_base_w
MAE: 2.34E-01
MSE: 2.63E-01
RMSE: 5.13E-01
Intercept: -1.45E-01
Slope: 9.09E-01
Coverage: 80.0%
Coverage target: 80.0%
log_base_w estimate
log_base_w truth
0
−1
−2
−3
−4
−5
−6
−7

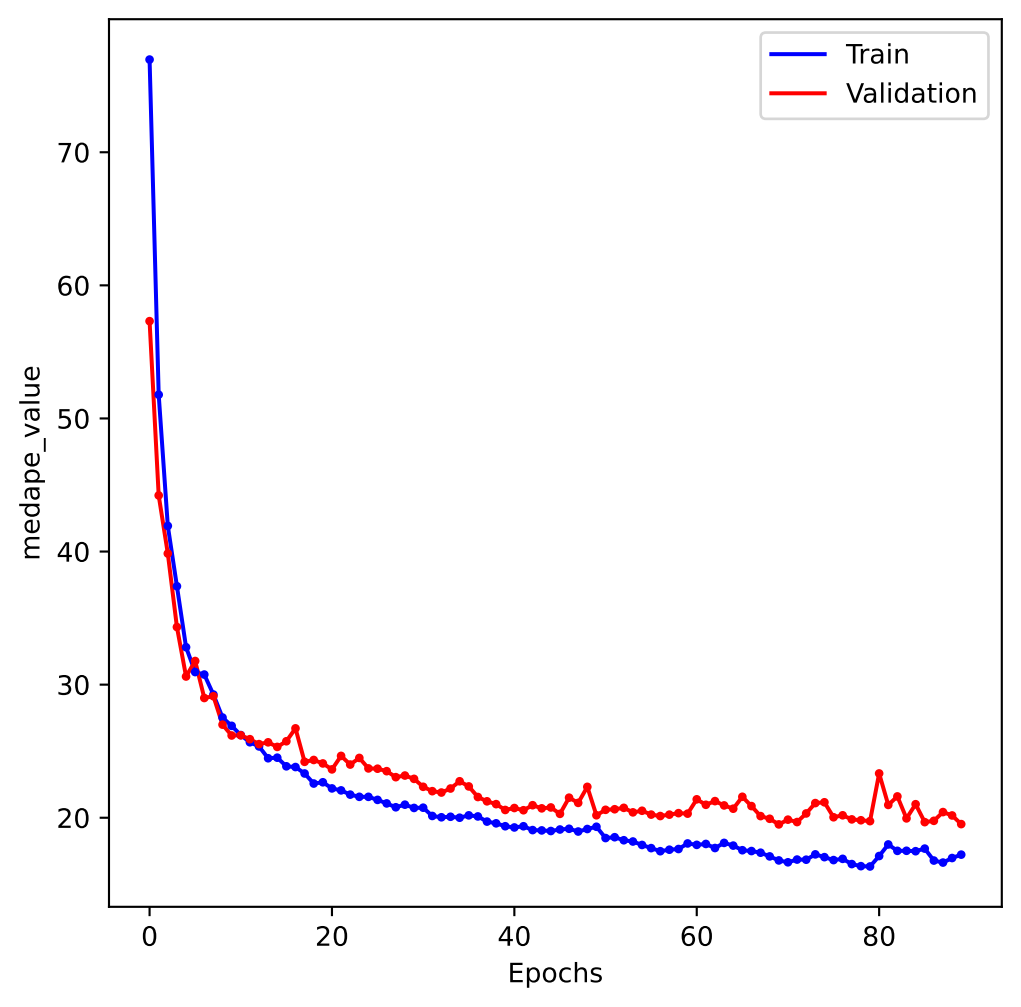

Train
Validation
loss_combined
Epochs
1.1
1.0
0.9
0.8
0.7
0.6
0.5
0.4
0
20
40
60
80

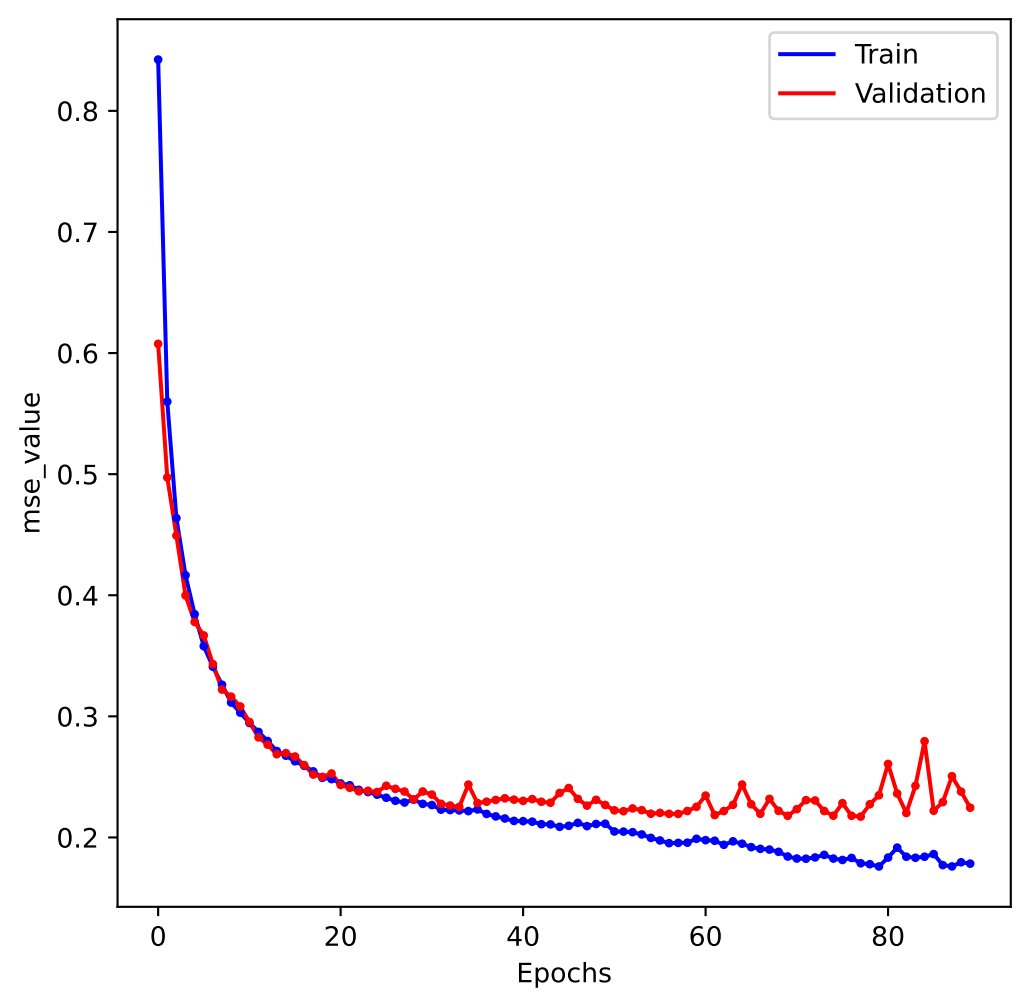

Train
Validation
loss_lower
0.5
0.4
0.3
0.2
0.1
0
20
40
60
80
Epochs

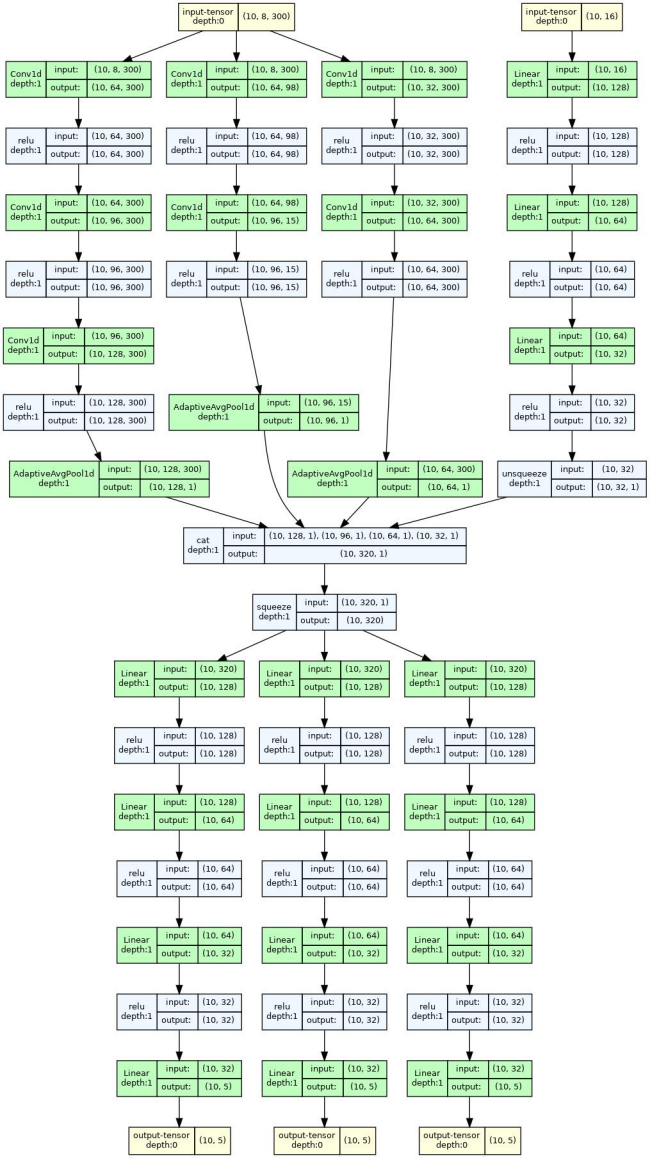

Train
Validation
loss_upper
0.18
0.16
0.14
0.12
0.10
0.08
0.06
0
20
40
60
80
Epochs


Train
Validation
loss_value
Epochs
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0
20
40
60
80


Train
Validation
0.6
0.5
0.4
0.3
mae_value
0
20
40
60
80
Epochs


Train
Validation
70
60
50
40
30
20
medape_value
0
20
40
60
80
Epochs


Train
Validation
mse_value
0.8
0.7
0.6
0.5
0.4
0.3
0.2
0
20
40
60
80
Epochs


input-tensor depth:0
(10, 8, 300)
input-tensor depth:0
(10, 16)
Conv1d depth:1
input: (10, 8, 300)
output: (10, 64, 300)
Conv1d depth:1
input: (10, 8, 300)
output: (10, 64, 98)
Conv1d depth:1
input: (10, 8, 300)
output: (10, 32, 300)
Linear depth:1
input: (10, 16)
output: (10, 128)
relu depth:1
input: (10, 64, 300)
output: (10, 64, 300)
relu depth:1
input: (10, 64, 98)
output: (10, 64, 98)
relu depth:1
input: (10, 32, 300)
output: (10, 32, 300)
relu depth:1
input: (10, 128)
output: (10, 128)
Conv1d depth:1
input: (10, 64, 300)
output: (10, 96, 300)
Conv1d depth:1
input: (10, 64, 98)
output: (10, 96, 15)
Conv1d depth:1
input: (10, 32, 300)
output: (10, 64, 300)
Linear depth:1
input: (10, 128)
output: (10, 64)
relu depth:1
input: (10, 96, 300)
output: (10, 96, 300)
relu depth:1
input: (10, 96, 15)
output: (10, 96, 15)
relu depth:1
input: (10, 64, 300)
output: (10, 64, 300)
relu depth:1
input: (10, 64)
output: (10, 64)
Conv1d depth:1
input: (10, 96, 300)
output: (10, 128, 300)
Linear depth:1
input: (10, 64)
output: (10, 32)
relu depth:1
input: (10, 128, 300)
output: (10, 128, 300)
AdaptiveAvgPool1d depth:1
input: (10, 96, 15)
output: (10, 96, 1)
relu depth:1
input: (10, 32)
output: (10, 32)
AdaptiveAvgPool1d depth:1
input: (10, 128, 300)
output: (10, 128, 1)
AdaptiveAvgPool1d depth:1
input: (10, 64, 300)
output: (10, 64, 1)
unsqueeze depth:1
input: (10, 32)
output: (10, 32, 1)
cat depth:1
input: (10, 128, 1), (10, 96, 1), (10, 64, 1), (10, 32, 1)
output: (10, 320, 1)
squeeze depth:1
input: (10, 320, 1)
output: (10, 320)
Linear depth:1
input: (10, 320)
output: (10, 128)
Linear depth:1
input: (10, 320)
output: (10, 128)
Linear depth:1
input: (10, 320)
output: (10, 128)
relu depth:1
input: (10, 128)
output: (10, 128)
relu depth:1
input: (10, 128)
output: (10, 128)
relu depth:1
input: (10, 128)
output: (10, 128)
Linear depth:1
input: (10, 128)
output: (10, 64)
Linear depth:1
input: (10, 128)
output: (10, 64)
Linear depth:1
input: (10, 128)
output: (10, 64)
relu depth:1
input: (10, 64)
output: (10, 64)
relu depth:1
input: (10, 64)
output: (10, 64)
relu depth:1
input: (10, 64)
output: (10, 64)
Linear depth:1
input: (10, 64)
output: (10, 32)
Linear depth:1
input: (10, 64)
output: (10, 32)
Linear depth:1
input: (10, 64)
output: (10, 32)
relu depth:1
input: (10, 32)
output: (10, 32)
relu depth:1
input: (10, 32)
output: (10, 32)
relu depth:1
input: (10, 32)
output: (10, 32)
Linear depth:1
input: (10, 32)
output: (10, 5)
Linear depth:1
input: (10, 32)
output: (10, 5)
Linear depth:1
input: (10, 32)
output: (10, 5)
output-tensor depth:0
(10, 5)
output-tensor depth:0
(10, 5)
output-tensor depth:0
(10, 5)